



The **Software Testing Life Cycle (STLC)** is a structured process that defines the sequence of activities performed during software testing to ensure quality and reliability.

It runs parallel to the **Software Development Life Cycle (SDLC)** and provides a systematic approach to validating software before release.



Software Testing Life Cycle (STLC)

STLC is the **framework of phases** that guide testing teams from requirement analysis to test closure. Each phase has **entry criteria, activities, exit criteria, and deliverables**, ensuring testing is consistent and measurable.



Phases of STLC

1. Requirement Analysis

- Review functional and non-functional requirements.
- Identify testable features and clarify ambiguities.

Deliverable: Requirement Traceability Matrix (RTM).

2. Test Planning

- Define scope, objectives, resources, schedule, and risk mitigation.

Deliverable: Test Plan document.

3. Test Case Development

- Design test cases, scripts, and prepare test data.

Deliverable: Test Case Repository.

4. Test Environment Setup

- Configure hardware/software to simulate production conditions.

Deliverable: Stable test environment.

5. Test Execution

- Run test cases, log defects, validate outcomes.

Deliverable: Defect reports, execution logs.

6. Test Cycle Closure

- Evaluate test coverage, defect density, lessons learned.

Deliverable: Test Summary Report.



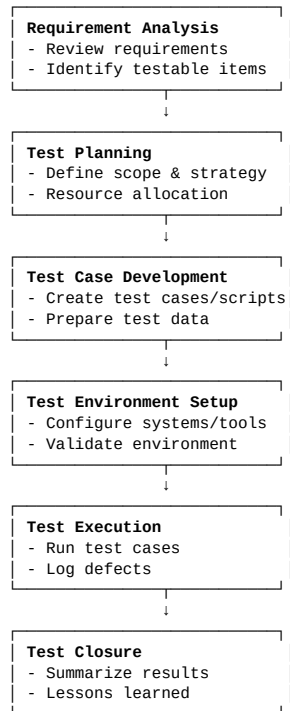
Quick Comparison Table

<u>Phase</u>	<u>Focus</u>	<u>Deliverable</u>
Requirement Analysis	Identify testable features	RTM
Test Planning	Define scope & strategy	Test Plan
Test Case Development	Create cases/scripts	Test Repository
Environment Setup	Configure systems	Test Environment
Execution	Run tests & log defects	Defect Reports
Closure	Evaluate & improve	Test Summary Report

- STLC ensures **early defect detection**, reducing cost and risk.
- It aligns testing with **business requirements** and **development phases**.
- In Agile/DevOps, these phases are **iterative and continuous** within each sprint or CI/CD pipeline.



The six phases of the Software Testing Life Cycle (STLC)



Phase Overview

- **Requirement Analysis** → Identify testable features.
- **Test Planning** → Define scope, resources, risks.
- **Test Case Development** → Build cases, scripts, data.
- **Test Environment Setup** → Prepare stable test systems.
- **Test Execution** → Run tests, log defects.
- **Test Closure** → Evaluate coverage, document lessons.

— This flow ensures testing is **systematic and measurable**, reducing risk and improving software quality. Each phase has **entry and exit criteria**, so teams know when to move forward.